Solar C appendix

Sun · C · downloadable appendix

Document	Solar C appendix
Subject	Solar photospheric abundance appendix · C
Author	Ryan Schmitt
Date	2026-10-18
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.202
Source	science commit 67a3cb4f0c66
Web	https://exoplanetcodex.org/systems/sol/elements/c/

Solar carbon: 8.507 ± 0.056

Asplund, Amarsi & Grevesse 2021: 8.46 ± 0.04 · this measurement +0.047 dex

Combined the way Asplund, Amarsi & Grevesse (2021) combine it: weighted mean and error of the indicator families [C I], C I, C2, CH, CO (Asplund+2021 Sect. 2.3). Each indicator is represented by our best analysis of it.

C I: 8.488 ± 0.069 · IAG FTS · red-optical · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

C2: 8.419 ± 0.210 · HARPS · VIS · C_2 Swan band (molecular band) · 1D LTE

CH: 8.509 ± 0.131 · HARPS · VIS · CH G-band (molecular band) · 1D LTE

CO: 8.701 ± 0.182 · CRIRES+ · K · MOL-CO_K · 1D LTE

Not measurable in our spectra: [C I].

Highlighted Products

VIS

8.430 ± 0.072

±0.052 stat · ±0.049 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

8.488 ± 0.069

±0.036 stat · ±0.059 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

NIR

8.559 ± 0.078

±0.011 stat · ±0.077 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

8.396 ± 0.174

±0.115 stat · ±0.131 syst

SET-ELGUETA26 · 1D LTE · CRIRES+ H (full arm, telluric-corrected) · Reference Grade

K

8.701 ± 0.182

±0.071 stat · ±0.168 syst

MOL-CO_K · 1D LTE · CRIRES+ · Codex Grade

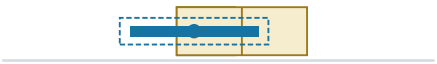
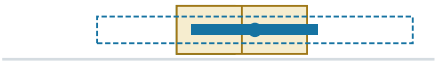
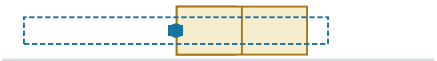
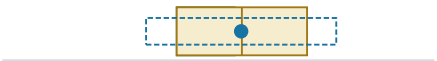
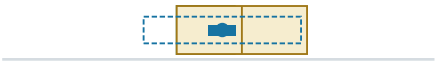
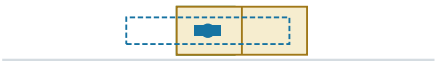
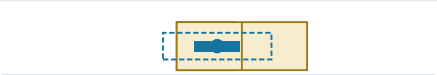
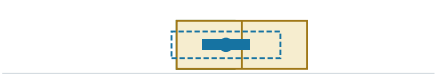
Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

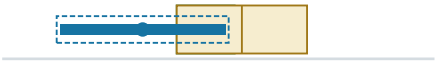
Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	29 products -- C I · Asplund et al. 2021 lines; SET-ELGUETA26
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	34 products -- C I 5052 Å (single line); C I 5380 Å (single line); C I · Codex-graded lines; CH G-band (molecular band); C_2 Swan band (molecular band); MOL-CO_K
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

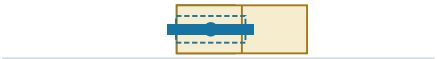
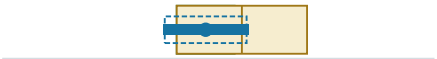
VIS — HARPS
A(C) dex; axis 8.18–8.78

C I 5052 Å (single line) · 1D LTE Codex Grade · 1 line		8.444 ±0.089 stat ±0.103 syst
C I 5380 Å (single line) · 1D LTE Codex Grade · 1 line		8.528 ±0.087 stat ±0.218 syst
C_2 Swan band (molecular band) · 1D LTE Codex Grade · 167 lines		8.419 ±0.010 stat ±0.210 syst
CH G-band (molecular band) · 1D LTE Codex Grade · 184 lines		8.509 ±0.006 stat ±0.131 syst
C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		8.483 ±0.019 stat ±0.109 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.463 ±0.019 stat ±0.113 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.476 ±0.019 stat ±0.113 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.476 ±0.031 stat ±0.075 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.488 ±0.033 stat ±0.075 syst

VIS — IAG FTS
A(C) dex; axis 8.18–8.78

C I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		8.373 ±0.114 stat ±0.119 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.355 ±0.115 stat ±0.123 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.367 ±0.113 stat ±0.122 syst

VIS — Kitt Peak (Kurucz 2005)
A(C) dex; axis 8.18–8.78

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		8.467 ±0.059 stat ±0.048 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.449 ±0.059 stat ±0.057 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.460 ±0.059 stat ±0.057 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.363 ±0.086 stat ±0.095 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.375 ±0.085 stat ±0.095 syst

VIS — Kitt Peak (1984, molecfit)
A(C) dex; axis 8.18–8.78

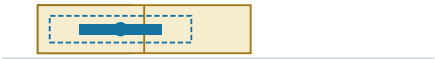
C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		8.450 ±0.052 stat ±0.038 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.430 ±0.052 stat ±0.049 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.443 ±0.052 stat ±0.049 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.376 ±0.085 stat ±0.082 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.388 ±0.084 stat ±0.082 syst

VIS axis: 8.18 · 8.33 · 8.48 · 8.63 · 8.78 A(C) dex

red-optical — IAG FTS
A(C) dex; axis 8.39–8.76

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		8.529 ±0.039 stat ±0.052 syst

red-optical — IAG FTS
A(C) dex; axis 8.39–8.76

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		8.488 ±0.036 stat ±0.059 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		8.490 ±0.035 stat ±0.060 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		8.566 ±0.037 stat ±0.080 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		8.528 ±0.040 stat ±0.085 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 6 lines		8.527 ±0.039 stat ±0.085 syst

red-optical — Kitt Peak (Kurucz 2005)
A(C) dex; axis 8.39–8.76

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		8.536 ±0.042 stat ±0.082 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		8.532 ±0.041 stat ±0.082 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		8.610 ±0.018 stat ±0.115 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		8.568 ±0.019 stat ±0.120 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 6 lines		8.563 ±0.018 stat ±0.119 syst

red-optical — Kitt Peak (1984, molecfit)
A(C) dex; axis 8.39–8.76

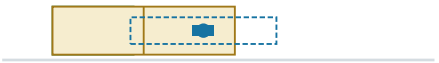
C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		8.587 ±0.046 stat ±0.048 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		8.541 ±0.043 stat ±0.055 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		8.540 ±0.042 stat ±0.057 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		8.607 ±0.026 stat ±0.075 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		8.572 ±0.027 stat ±0.080 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 6 lines		8.573 ±0.027 stat ±0.080 syst

red-optical axis: 8.39 · 8.48 · 8.57 · 8.66 · 8.76 A(C) dex

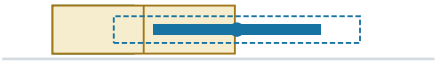
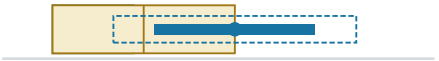
NIR — CRIRES+ J
A(C) dex; axis 8.37–8.80

SET-ELGUETA26 · 1D LTE Reference Grade · 12 lines		8.488 ±0.022 stat ±0.088 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		8.521 ±0.018 stat ±0.070 syst

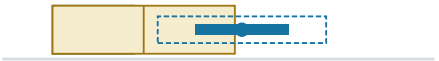
NIR — CRIRES+ Y (full arm)
A(C) dex; axis 8.37–8.80

SET-ELGUETA26 · 1D LTE Reference Grade · 6 lines		8.569 ±0.010 stat ±0.072 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		8.580 ±0.014 stat ±0.118 syst

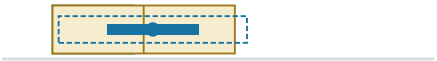
NIR — IAG FTS
A(C) dex; axis 8.37–8.80

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		8.677 ±0.066 stat ±0.090 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.602 ±0.083 stat ±0.122 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.600 ±0.079 stat ±0.120 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		8.677 ±0.066 stat ±0.090 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		8.602 ±0.083 stat ±0.122 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		8.600 ±0.079 stat ±0.120 syst

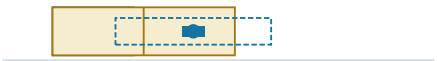
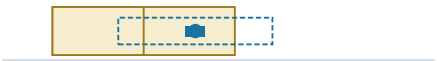
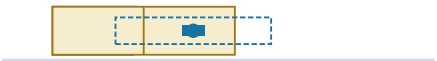
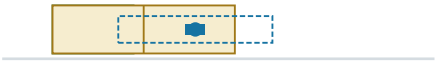
NIR — Kitt Peak (Kurucz 2005)
A(C) dex; axis 8.37–8.80

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		8.607 ±0.046 stat ±0.083 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.519 ±0.045 stat ±0.093 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.522 ±0.043 stat ±0.093 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		8.607 ±0.046 stat ±0.083 syst

NIR — Kitt Peak (Kurucz 2005)
A(C) dex; axis 8.37–8.80

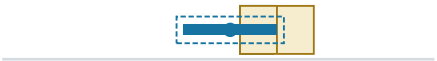
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		8.519 ±0.045 stat ±0.093 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		8.522 ±0.043 stat ±0.093 syst

NIR — Kitt Peak (1984, molecfit)
A(C) dex; axis 8.37–8.80

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		8.623 ±0.011 stat ±0.097 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.559 ±0.011 stat ±0.077 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.561 ±0.010 stat ±0.076 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		8.559 ±0.011 stat ±0.077 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		8.561 ±0.010 stat ±0.076 syst

NIR axis: 8.37 · 8.48 · 8.58 · 8.69 · 8.80 A(C) dex

H — CRIRES+ H (full arm, telluric-corrected)
A(C) dex; axis 7.84–8.89

SET-ELGUETA26 · 1D LTE Reference Grade · 4 lines		8.396 ±0.115 stat ±0.131 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		8.367 ±0.157 stat ±0.497 syst

H axis: 7.84 · 8.10 · 8.37 · 8.63 · 8.89 A(C) dex

K — CRIRES+
A(C) dex; axis 8.39–8.90

MOL-CO_K · 1D LTE Codex Grade · 64 lines		8.701 ±0.071 stat ±0.168 syst

K axis: 8.39 · 8.52 · 8.64 · 8.77 · 8.90 A(C) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every carbon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the carbon abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C_2, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(C)	Largest uncertainty terms (dex)
VIS	C I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2	8.355–8.476 (3D-NLTE)	continuum placement 0.05, line strengths (gf) 0.04
VIS	C I 5052 Å (single line)	HARPS	1D-LTE	1	8.444 (1D-LTE)	continuum placement 0.09, fit / line-to-line scatter 0.09
VIS	C I 5380 Å (single line)	HARPS	1D-LTE	1	8.528 (1D-LTE)	continuum placement 0.15, spread between spectra 0.12
VIS	C_2 Swan band (molecular band)	HARPS	1D-LTE	167	8.419 (1D-LTE)	line strengths (gf) 0.13, microturbulence 0.12
VIS	CH G-band (molecular band)	HARPS	1D-LTE	184	8.509 (1D-LTE)	line strengths (gf) 0.11, spread between spectra 0.06
VIS	C I · Asplund et al. 2021 lines	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	3	8.430–8.463 (3D-NLTE)	fit / line-to-line scatter 0.05, line strengths (gf) 0.03
red-optical	C I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	6	8.528–8.572 (3D-NLTE)	fit window 0.05, model atmosphere 0.04
red-optical	C I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	10	8.488–8.541 (3D-NLTE)	fit / line-to-line scatter 0.04, model atmosphere 0.03
NIR	C I · Codex-graded lines	CRIRES+ J, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2–4	8.519–8.602 (3D-NLTE)	telluric residual 0.04, fit window 0.04
NIR	C I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	3–4	8.519–8.602 (3D-NLTE)	telluric residual 0.04, fit window 0.04
NIR	SET-ELGUETA26	CRIRES+ J, CRIRES+ Y (full arm)	1D-LTE	6–12	8.488–8.569 (1D-LTE)	telluric residual 0.05, fit window 0.05

Band	What was measured	Solar spectra	Treatments	Lines	Result A(C)	Largest uncertainty terms (dex)
H	C I · Codex-graded lines	CRIRES+ H (full arm, telluric-corrected)	1D-LTE	3	8.367 (1D-LTE)	telluric residual 0.44, fit window 0.21
H	SET-ELGUETA26	CRIRES+ H (full arm, telluric-corrected)	1D-LTE	4	8.396 (1D-LTE)	fit / line-to-line scatter 0.11, telluric residual 0.11
K	MOL-CO_K	CRIRES+	1D-LTE	64	8.701 (1D-LTE)	model atmosphere 0.14, coupling to the other C/N/O abundances 0.08

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 1 lines.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
C I	5380.340	saturated

Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, C_2 and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

C/O indicator strategy · Methodology

Download the evidence

Download PDF — Solar C appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 67a3cb4f0c66b8719fb12fe68fa9d992503d35eb
Feed C.json v1.202 · updated 2026-10-10T22:00:32Z
Generated 2026-10-18T00:15:00-06:00 (source commit timestamp)
Feed SHA-256 f0fae34963f2961157ebbdc444c49f12da561063e7bc980547cba6fc3131066a
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

Atomic data and gf authorities. AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign’s NIST/MCHF adjudication; no gf source is substituted here. Sources: [5].

Model and grid sources. Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [6].

Atomic data and gf authorities. Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

Molecular line data and delivery. CO line-data delivery: CO ExoMol line list, Li et al. 2015 (ApJS 216, 15). RYA-236 conversion of the ExoMol Li2015 CO list to the Turbospectrum babsma .dat format (converted to species code 0608.012016; conversion script external to this repo — RYA-236) ExoMol/Turbospectrum is a redistribution/conversion, not the primary publication. Sources: [2], [7], [8].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [6].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [6].

Molecular line data and delivery. C2 line-data delivery: C2 Swan $d^3\Pi g–a^3\Pi u$ (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [9].

Molecular line data and delivery. CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest’s MNRAS locator conflicts with that record. Sources: [2], [7], [10].

Product-to-source index

H · CRIRES+ H (full arm, telluric-corrected) · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7]
C-products-f04678f420c3

H · CRIRES+ H (full arm, telluric-corrected) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7]
C-products-7015fc1e2bcc

K · CRIRES+ · MOL-CO_K · 1D LTE · [1], [2], [7], [8] C-products-a3286d05d53c

NIR · CRIRES+ J · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-products-9b721d3f7ec5

NIR · CRIRES+ J · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-ed9b75aee66f

NIR · CRIRES+ Y (full arm) · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-products-6449cb5086f8

NIR · CRIRES+ Y (full arm) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-52eb6ae4e11b

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-65dcaa1cf605

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-b80d1c35e7d3

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-fe537374f04a

NIR · IAG FTS · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-91b412d5a9bb

NIR · IAG FTS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-fe0de6425334

NIR · IAG FTS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-f2e0374fd8bc

NIR · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-7d51b738cffa

NIR · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-0e050cb3a6e5

NIR · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-107337367f87

NIR · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-0783b41c5813

NIR · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-a78222e96405

NIR · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-8585afef7622

NIR · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-d60eacfa6f91

NIR · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-d771b79962fd

NIR · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-318e38e4371d

NIR · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-9dca3eb1160a

NIR · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-d66637bbc168

VIS · HARPS · C I 5052 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-7fc96af7bda6

VIS · HARPS · C I 5380 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-536db2a5fb17

VIS · HARPS · C_2 Swan band (molecular band) · 1D LTE · [1], [2], [7], [9] C-products-4309ba5d0384

VIS · HARPS · CH G-band (molecular band) · 1D LTE · [1], [2], [7], [10] C-products-19d90d71f47b

VIS · HARPS · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-869c2e97dcc1

VIS · HARPS · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-50838ebd563e

VIS · HARPS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-31d1446fd624

VIS · HARPS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-9a0633f65045

VIS · HARPS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-04337f1543a3

VIS · IAG FTS · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-9e24685f5277

VIS · IAG FTS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-4c4864fafbe3

VIS · IAG FTS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-827b1b7594ef

VIS · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-b78395f299d8

VIS · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-dbd5924ece83

VIS · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-c4c75e2631fd

VIS · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-4abf4b2eb643

VIS · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-f50ea16aadda

VIS · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-cc2cf7c54bd1

VIS · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-2f58bf34b212

VIS · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-1bb009464841

VIS · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-43cf7156ceaf

VIS · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-8976459a47b2

red-optical · IAG FTS · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-64ce4eab7a9a

red-optical · IAG FTS · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-48116a3becd2

red-optical · IAG FTS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-312499053699

red-optical · IAG FTS · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-0e2f97eba4d6

red-optical · IAG FTS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-1b70c80db983

red-optical · IAG FTS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-226d6a8985d7

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-11b953d2548e

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-ca783ee3d2a1

red-optical · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-68a3b821aad6

red-optical · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-a75d43a2df98

red-optical · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-39d9eb4dae98

red-optical · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-ef56049f28cd

red-optical · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-27245f742fdd

red-optical · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-8ee6dec61b1a

red-optical · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-0fee360c4682

red-optical · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-ef9ba82337b9

red-optical · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-f49fa3470e17

VIS · HARPS · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-quarantine-450b744f3f97

VIS · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-quarantine-2f69a108b1e7

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-quarantine-bdee5b90296a

NIR · Kitt Peak (1984, molecfit) · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-quarantine-d13227a59f31

NIR · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-quarantine-0121df2035b5

VIS · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-quarantine-9b3b9069e4e7

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. source record

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

[5] — Li, Amarsi, Papoulia, Ekman & Jonsson 2021, MNRAS 502, 3780 DOI: 10.1093/mnras/stab214. source record

Model and grid sources

[6] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. source record

[7] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

Molecular line data and delivery

[8] — Li et al. 2015, ApJS 216, 15 DOI: 10.1088/0067-0049/216/1/15. source record

[9] — Brooke et al. 2013, JQSRT 124, 11 DOI: 10.1016/j.jqsrt.2013.02.025. source record

[10] — Masseron et al. 2014, A&A 571 A47 DOI: 10.1051/0004-6361/201423956. source record